Per-ROI hierarchical Bayesian GAM
(projection away from HP vs distance from HP)

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WHAT IS FIT (one model PER ROI — no joint master model):
  proj ~ hsgp(distance, m=8, c=1.5) + (1 | subject)

  • `hsgp(...)` is bambi's Hilbert-space Gaussian-process approximation,
    the equivalent of `s(distance, k=8)` in mgcv.
  • `(1 | subject)` = per-subject random intercept.
  • family = "gaussian".
  • HSGP length-scale prior:
      ell ~ LogNormal(mu=0.693 = log(2.00°),
                      sigma=0.50)
    (slightly informative, centered around ~2.0°).
    HSGP marginal SD: sigma ~ Exponential(1.0).
  • Sampler: NUTS via PyMC, draws = 2000, tune = 2000,
    chains = 4, target_accept = 0.99.

DATA PIPELINE (same as plot_projection_clusters.py):
  • load_all_conditionwise(bids_folder='/data/ds-retsupp', r2_thr=0.2).
  • filter_prf_inside_aperture(aperture_radius=3.17,
      max_fraction_outside=0.5,
      use_mean_only=False).
  • Outlier clip: drop voxel-condition rows with
      |proj_away_from_HP| > 1.5°  (proj = -dy_rotated).
  • Per (subject, ROI, distance-bin): mean of proj across voxel-condition
    rows in the cell. Distance bin width = 0.5°,
    centers from 0.25° to
    6.75°.

POSTERIOR SUMMARY:
  • Smooth f(distance) is evaluated at 50 points from
    0.50° to 6.50°. Per distance:
      - posterior median + 95% HDI of the population-level mean.
      - P(f > 0): probability the smooth is > 0 at that distance.
      - P(f < 0): probability the smooth is < 0 at that distance.
  • A distance is flagged ("Bayesian cluster") when
      P(>0) > 0.95  →  red shading
      P(<0) > 0.95  →  blue shading

NOTES:
  • Each ROI = a separate bambi model + separate posterior trace.
  • Sampler warnings (divergences, etc.) are printed per-ROI.
  • Default seed = 42; use --seed to vary.

ROIs fit: ['V1', 'V2', 'V3', 'V3AB', 'hV4', 'LO', 'TO', 'VO']
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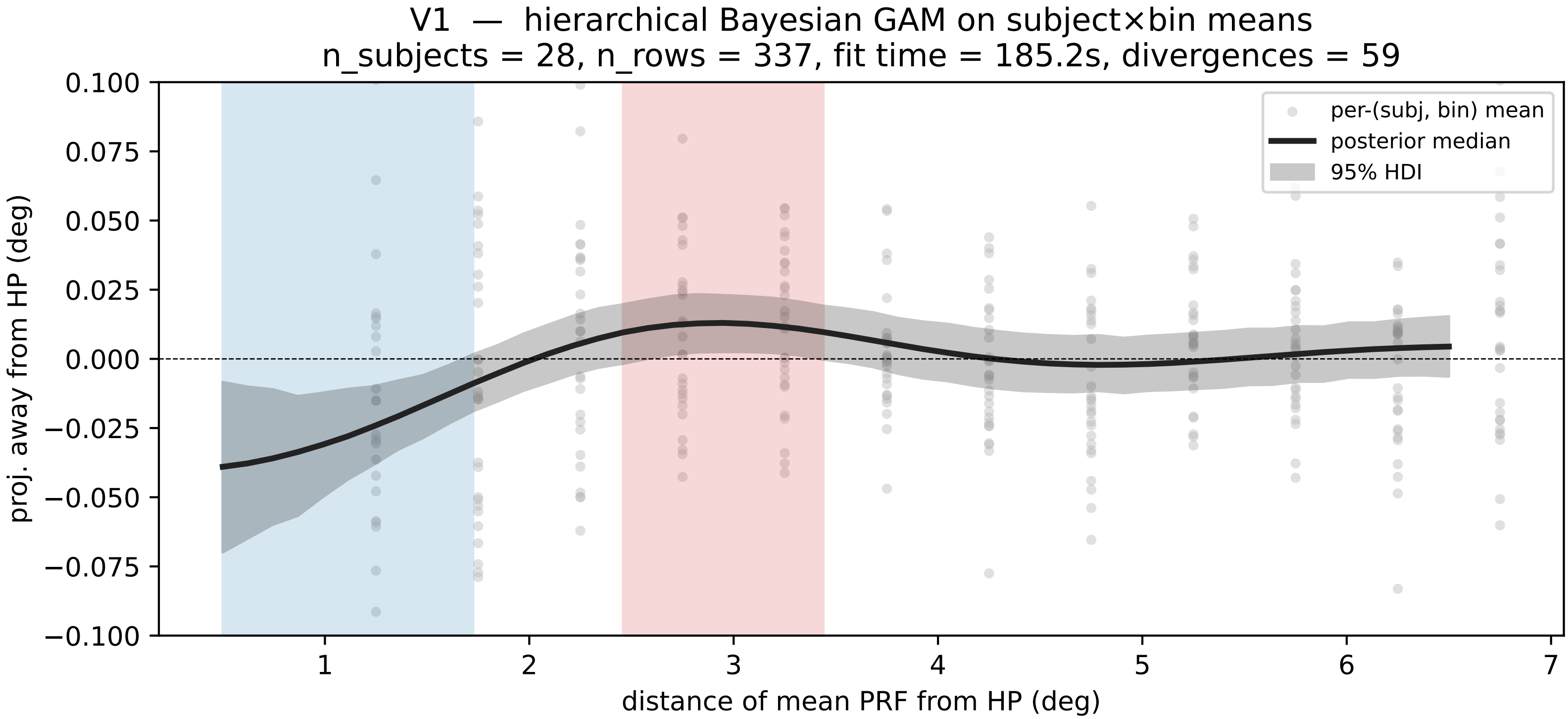
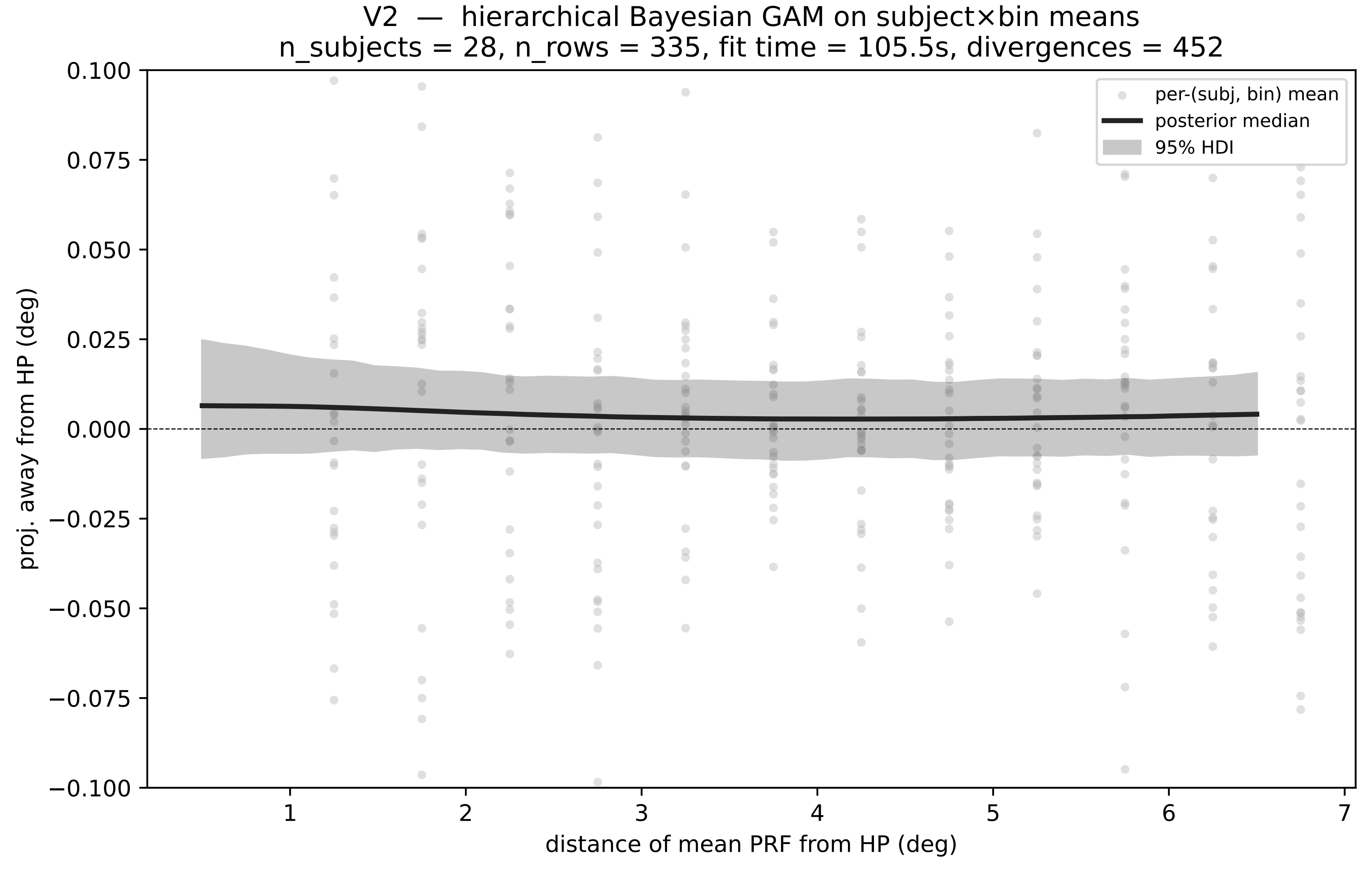
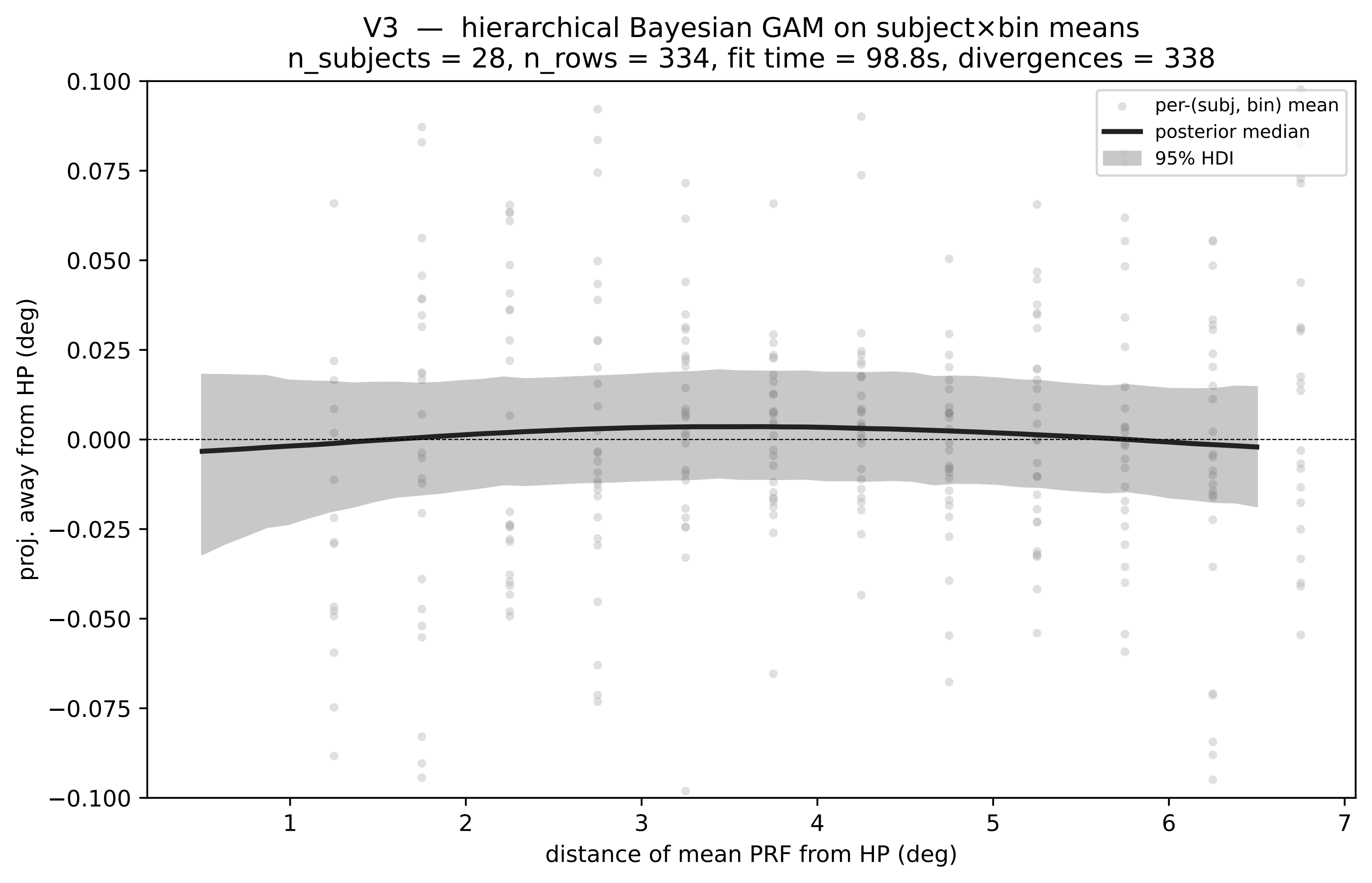
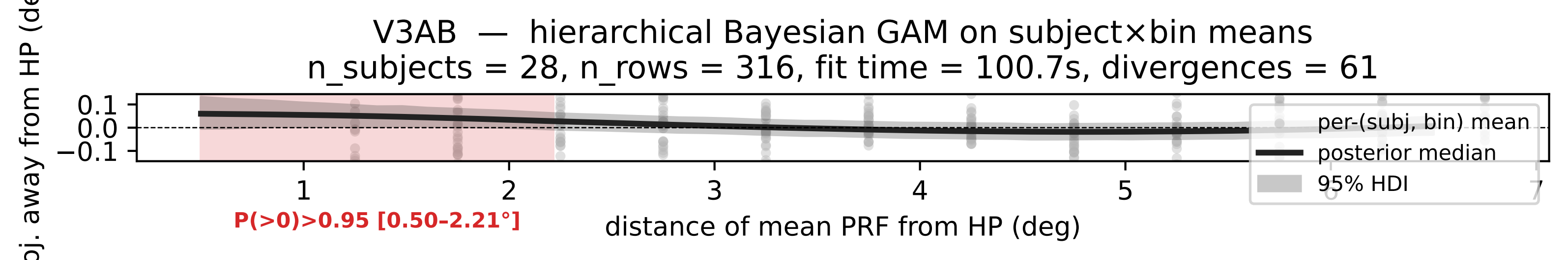


P(<0)>0.95 [0.50–1.72°]   P(>0)>0.95 [2.46–3.44°]

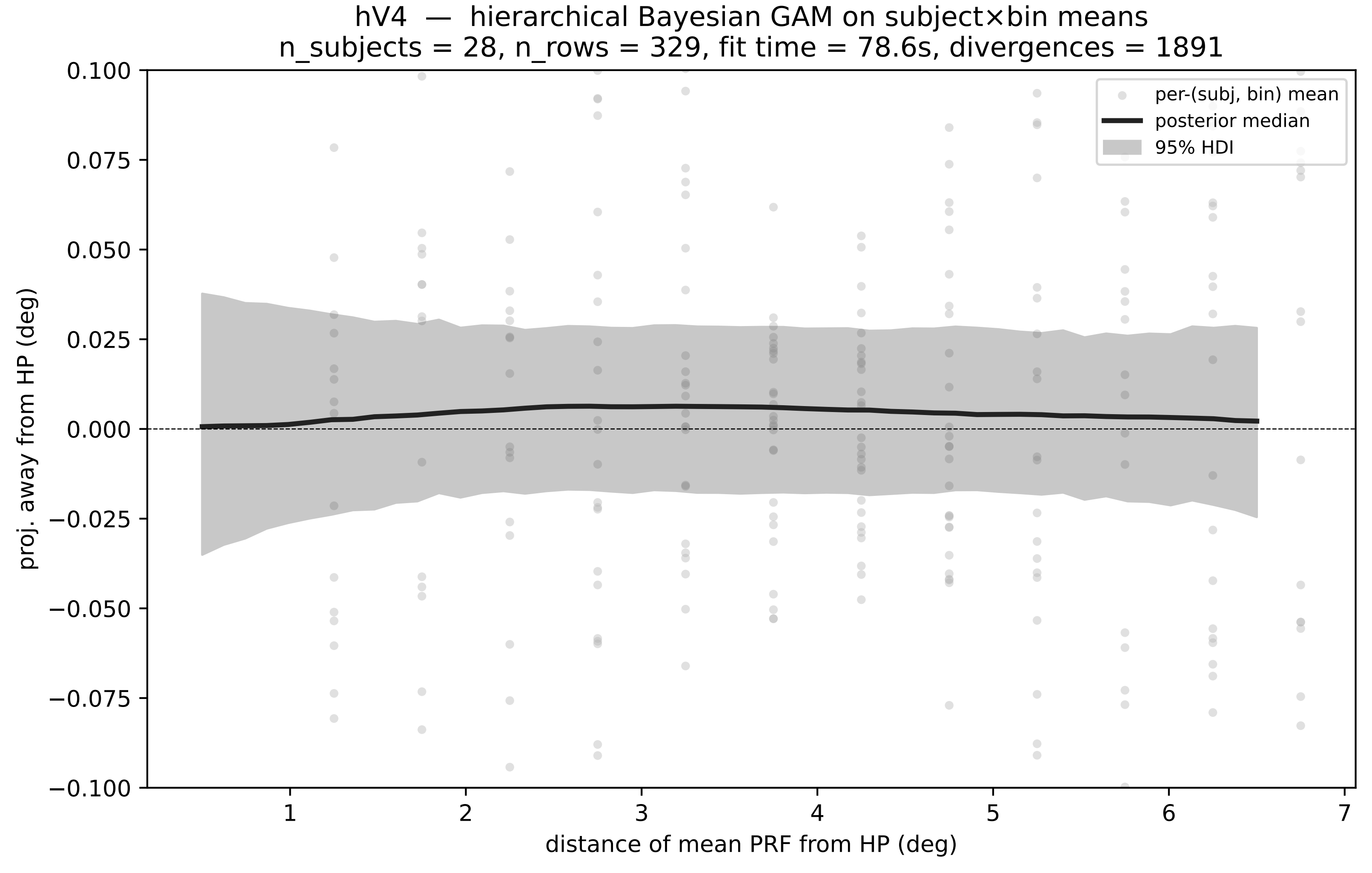

V2 — hierarchical Bayesian GAM on subject×bin means
n_subjects = 28, n_rows = 335, fit time = 105.5s, divergences = 452
per-(subj, bin) mean
posterior median
95% HDI
proj. away from HP (deg)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
1
2
3
4
5
6
7
distance of mean PRF from HP (deg)



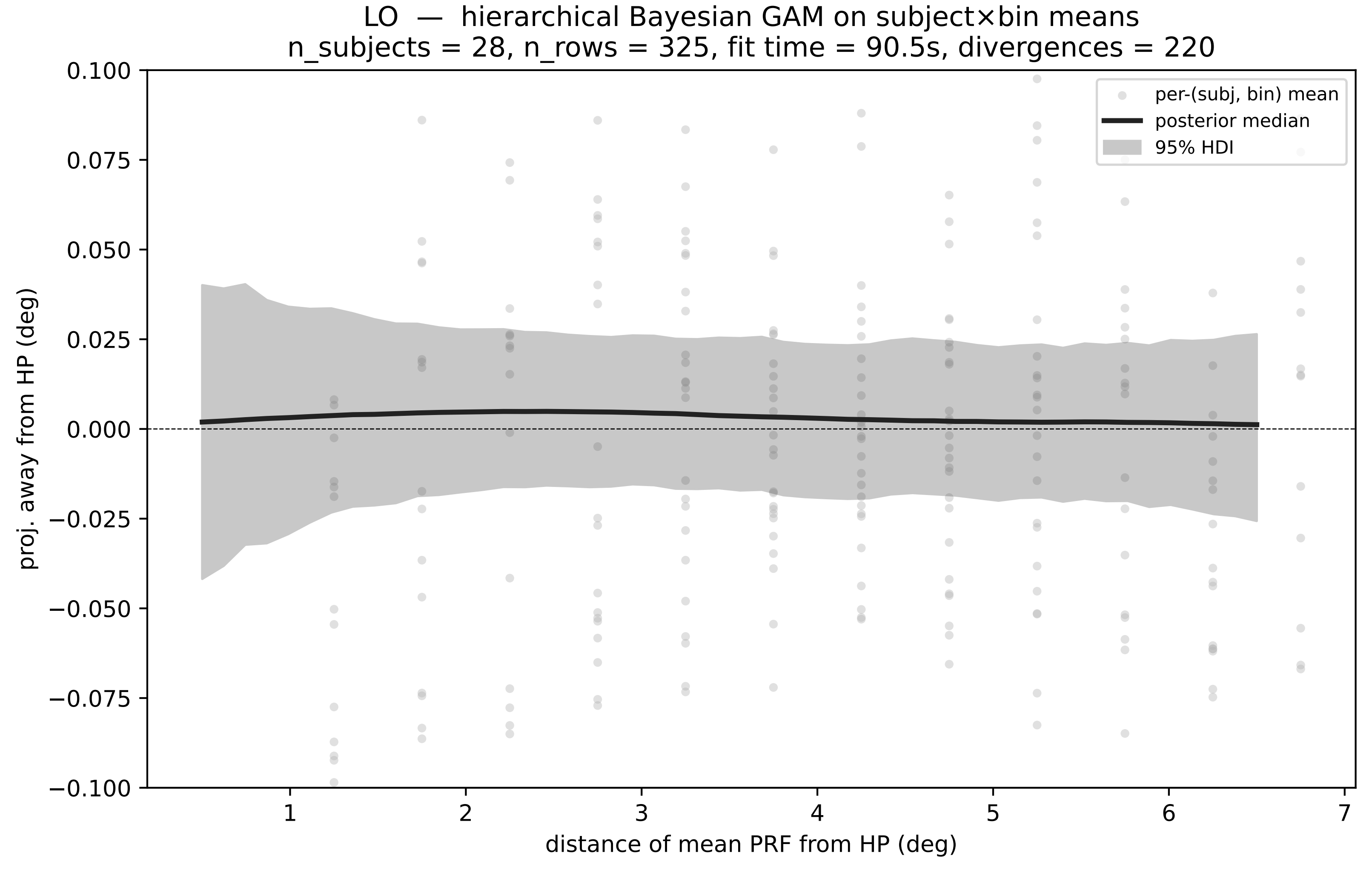
V3 — hierarchical Bayesian GAM on subject×bin means
n_subjects = 28, n_rows = 334, fit time = 98.8s, divergences = 338
per-(subj, bin) mean
posterior median
95% HDI
proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
1
2
3
4
5
6
7



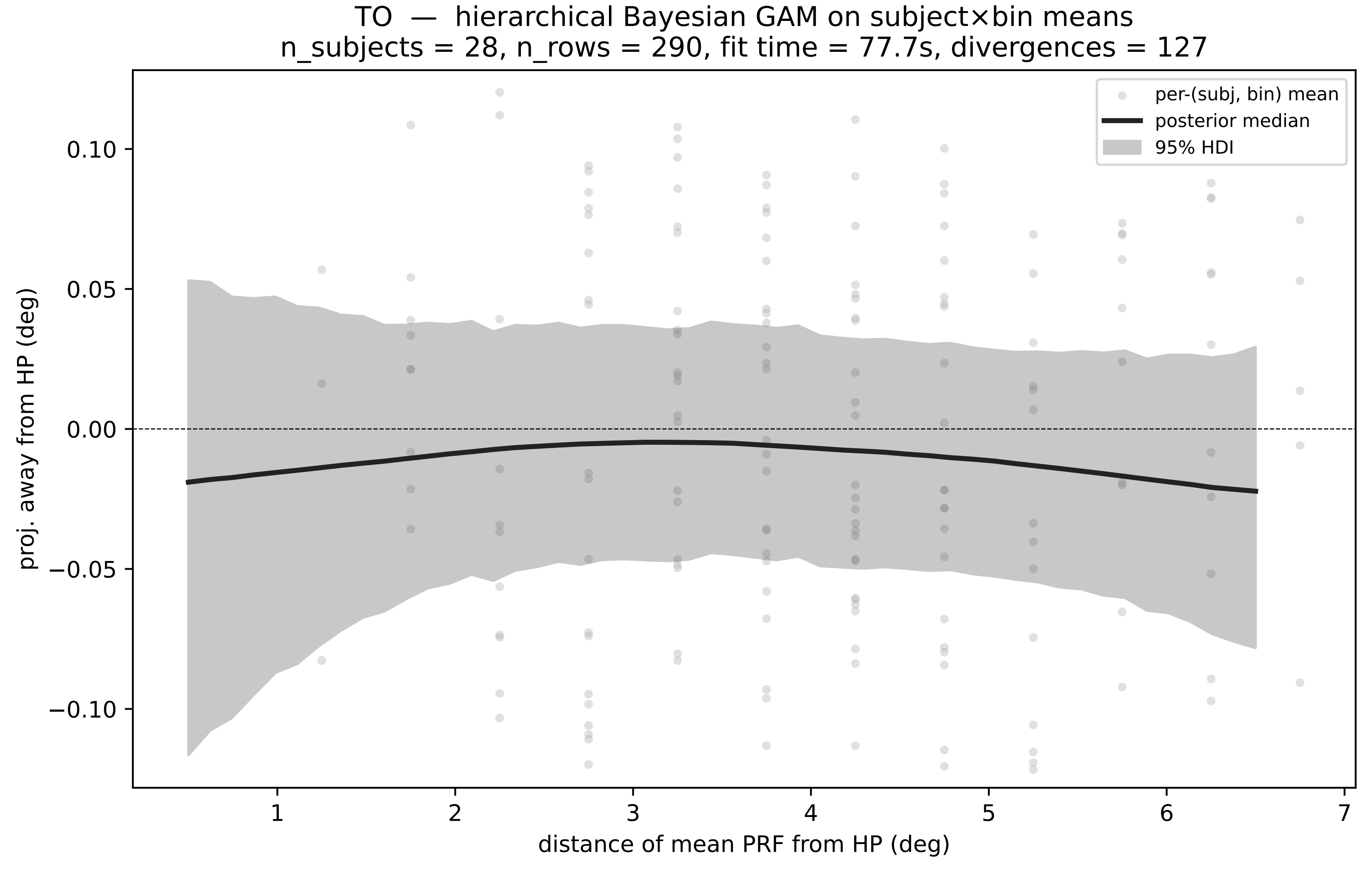
V3AB — hierarchical Bayesian GAM on subject×bin means
n_subjects = 28, n_rows = 316, fit time = 100.7s, divergences = 61
0.1
0.0
−0.1
1
2
3
4
5
6
7
per-(subj, bin) mean
posterior median
95% HDI
P(>0)>0.95 [0.50–2.21°]
distance of mean PRF from HP (deg)
oj. away from HP (de



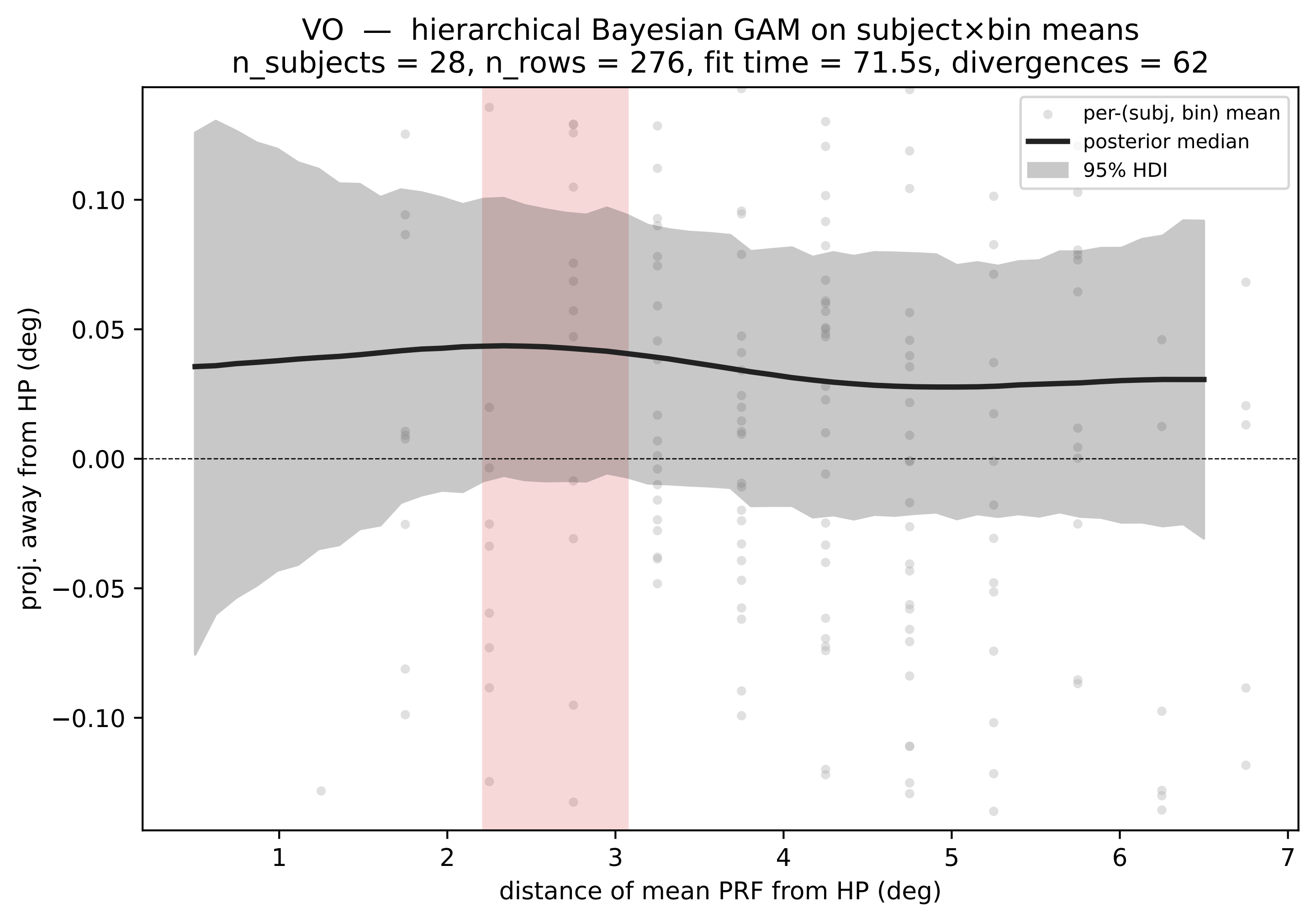
hV4 — hierarchical Bayesian GAM on subject×bin means
n_subjects = 28, n_rows = 329, fit time = 78.6s, divergences = 1891
per-(subj, bin) mean
posterior median
95% HDI
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
proj. away from HP (deg)
1
2
3
4
5
6
7
distance of mean PRF from HP (deg)



LO — hierarchical Bayesian GAM on subject×bin means
n_subjects = 28, n_rows = 325, fit time = 90.5s, divergences = 220
per-(subj, bin) mean
posterior median
95% HDI
proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
−0.075
−0.100
1
2
3
4
5
6
7



TO — hierarchical Bayesian GAM on subject×bin means
n_subjects = 28, n_rows = 290, fit time = 77.7s, divergences = 127
per-(subj, bin) mean
posterior median
95% HDI
proj. away from HP (deg)
distance of mean PRF from HP (deg)
0.10
0.05
0.00
−0.05
−0.10
1
2
3
4
5
6
7



P(>0)>0.95 [2.21–3.07°]